



Numerical simulation of the single-phase immersion cooling process using a dielectric fluid in a data server

Aditya Chhetri^a, Devendra Kashyap^a, Arvind Mali^a, Chaitanya Agarwal^a, Caroline Ponraj^b, N. Gobinath^{a,*}

^a School of Mechanical Engineering, Vellore Institute of Technology Chennai, Tamil Nadu 600127, India

^b School of Advanced Sciences, Vellore Institute of Technology Chennai, Tamil Nadu 600127, India

ARTICLE INFO

Article history:

Available online 7 November 2021

Keywords:

CFD
Immersion cooling
Dielectrics
Data centers
Power density

ABSTRACT

In the wake of novel technologies such as artificial intelligence (AI), machine learning (ML), internet of things (IoT) to cater to a wide spectrum of industrial applications, eventually the demand for ultra-fast processors/data centers has been on rise. In further, these novel technologies inherently result a drastic increase of power density in data centers, where air-cooling is the popularly utilized method to handle the thermal loads. The present research study involves in numerically investigating the effectiveness of single-phase immersion cooling process, an alternate solution to air-cooling method, on governing the heat load of the densely packed servers. A dielectric fluid, EC-110 is chosen for the simulation on a suitably developed CAD model of a server, of rated design power of 150 W. Four different processing unit powers of value 65 W, 90 W, 115 W and 150 W are supplied as input load to the cooling fluid in the simulation work. The condition of inlet temperature of the dielectric fluid is varied from 25°C to 40°C and the flow rates are varied from 0.5 lpm to 2.5 lpm. The resulting chip junction temperature in the server is estimated for different flow rates and inlet temperature conditions of the fluid, under immersion cooling technique. It proves that this cooling technique has the potential to ensure safe operation temperature of the server at low flow rates of the fluid with high inlet temperature, at high power densities.

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Selection and peer-review under responsibility of the scientific committee of the Second International Conference on Sustainable Energy Solutions for a Better Tomorrow.

1. Introduction

A data center is a huge facility that houses a cluster of networked computer servers that are often used by organizations to remotely store, analyze, or transfer massive quantities of data. With the rise of technologies like Machine Learning (ML), Internet of Things (IoT), and Artificial Intelligence (AI), the power density of servers has risen exponentially. This, in turn, necessitates the deployment of efficient cooling systems to ensure proper functioning of data centers. Air cooling has been used traditionally till date for cooling of data centers. In this method, the servers are arranged according to hot and cold aisles to avoid mixing of hot and cold air. This is a costly method and cooling alone accounts for one-third of the overall power usage as reported by Gandhi et. al [1]. Liquid cooling, which uses water, oil, and synthetic fluids as the coolant, has the capability to remove higher amount of heat from densely packed servers. Depending on the contact between the fluid and

server components, this approach is categorized into indirect cooling (Direct-to-Chip Cooling) and direct cooling (Immersion Cooling). Gandhi et. al [1] developed a CFD model of an HP ProLiant DL160 G6 1U server in 6SigmaET, and tested the performance of Mineral Oil and a synthetic dielectric fluid EC-100. They found that even when the inlet temperature of oil is 45 °C, the chip junction temperature is maintained at around 70 °C which is quite safe. In indirect cooling, non-dielectric fluids (water and glycol) are circulated through cold plates that are directly connected to the heat-generating components, avoiding direct contact between the electronic components and the cooling fluid [2]. Direct cooling refers to a system in which the servers transfer heat to cooling fluids through direct contact. Immersion cooling is further classified into single-phase and two-phase immersion cooling. In single phase immersion cooling, the servers are submerged in a sealed container containing the fluid through which heat transmission occurs. The cooling fluid stays in the liquid phase throughout the cooling process. The hot fluid is then pushed to a heat exchanger before being recirculated back into the tank. On the other hand, in two-phase immersion cooling, the fluid evaporates after absorbing heat from

* Corresponding author.

E-mail address: gobinath.n@vit.ac.in (N. Gobinath).

the chips and condenses with the assistance of a heat exchanger. Due to latent heat, the heat removal capability for two-phase immersion cooling is higher. The servers, like in single phase cooling, are immersed in a sealed container that may be readily removed. The advantages of dielectric liquids are that they possess high dielectric strength, good heat transfer characteristics and chemically stable nature under the range of conditions the system operates. Dielectric strength is the voltage that an insulating material can withstand before breakdown occurs. Generally, the dielectric strength of liquid is 10^7 V/cm times greater than air. Liquid coolants are found to show better ability to transfer heat and heat density than air, and henceforth, it can outperform air as a cooling medium. The critical factors behind the selection of the electro cool-110 are; its biodegradability and stability that makes EC-110 more attractive and an excellent option compared to other liquid coolants.

The servers of the data centers often operate continuously, and have to be maintained below safe operating temperatures to avoid its failure. The maximum allowable chip junction temperature is to be around 85 °C to avoid malfunctions, as discussed by Alfonso and Giulio [3]. Liquid immersion cooling demands no fans and hence the air handling unit is replaced by a pump, causing significant reduction in power requirements relative to conventional air cooling methods. Additionally, the space required for immersion cooling technology is reduced to almost one-third of the space required for conventional air cooling technologies which is mainly because of its higher Power Usage Effectiveness and rack density over the conventional cold plate and air-cooling technologies [4]. Cheng et. al [5] designed a single-phase immersion cooling system using 3 M Novec 7100 as the cooling fluid for an Intel i9 based CPU of a personal computer, also performing a CFD analysis of the same. The authors found no significant difference between the performances of a Copper Heat Sink and an Aluminium Heat Sink. An et. al [6] developed a computational model of a two-phase cooling system in ANSYS Fluent. The arrangement consisted of chips, modelled as a 5 cm × 5 cm square, mounted on vertical servers, immersed in a tank containing 3M Novec 7000, the phase change coolant. It was found that even at high powers on 200 W and above, the junction temperature could still be maintained below safe limits. Eiland et. al [7] developed an experimental setup of an immersion cooled Open Compute server, to explore the flow rates and inlet temperatures necessary for circulating the coolant (mineral oil). The authors also calculated the partial PUE of the setup, which was found to be as low as 1.03. The ideal flow rates for the oil were found to be between 0.5 and 2.5 lpm, with best efficiency obtained at lower flow rates and higher inlet temperatures. Shah et. al [8] used ANSYS Icepack to develop a CFD model of a 3rd Generation Open Compute Server to demonstrate thermal shadowing, and also optimised the heat sink to reduce its volume by 5 times. While there was a slight increase in the maximum chip junction temperature, it was within the safe limits. Shah et. al [9] studied the impact of form factor on the performance of dielectric fluid EC-100 for a constant power of 65 W and formulated the results. They found that the maximum chip junction temperatures increased as the form factor was reduced, and that EC-100 performed slightly better than mineral oil in cooling the servers.

Existing research works focused on servers with a form factor of 2U. However, a dense packing of the servers with form factor of <2U would facilitate better space utilization and hence helps to establish packing servers of form factors 1U and 1.5U. Besides, much research has been conducted for a fixed value of the power and no data is available on the temperature distribution on such servers at higher values of CPU power. This research work involves the use of a new cooling fluid, EC-110, and a conjugate heat transfer analysis on commercially used 3rd Generation Open Compute Server with modified form factor (reduced from 2U to 1 Open Rack

Unit) on ANSYS Fluent. To understand the fluid effectiveness, the analysis is carried out at different values for the input parameters such as power, inlet temperature, and flow rate of the fluid. Additionally, a predictive regression model is developed with the chip junction temperature as the output.

2. Methodology

This study consists of a conjugate heat transfer CFD analysis of a commercially used server [8] which is modified in order to facilitate a denser packing. It begins with a thorough literature study of the prevalent methods of cooling of data centers. Immersion Cooling is found to be the emerging cooling method with huge scope of development in future. The next step is to find a suitable cooling fluid. EC-110, a dielectric fluid, is selected, based on the properties (Specific Heat). A CAD model of a commercially used server is developed using SolidWorks, and transferred to ANSYS Fluent. A grid independence study is performed to find the optimum count for the number of elements and nodes. Validation of the model is done using previously published data with a different fluid. Then, the CFD analysis is performed at four different values of the power namely 65 W, 90 W, 115 W, and 150 W, four different values of inlet temperature namely 25°C, 30°C, 35°C, and 40°C for each value of the power, and at 5 different flow rates of the cooling fluid namely 0.5 lpm, 1 lpm, 1.5 lpm, 2 lpm, and 2.5 lpm for each value of inlet temperature. The results are tabulated, and a regression model is developed with the output as the chip junction temperature and the inputs as the power, flow rate and inlet temperature.

2.1. Cooling fluid

Manufactured by Engineered Fluids [10], Electrocool-110 is chemically a hydrocarbon with a paraffinic structure. It is non-polar, electrically non-conductive, and biodegradable. A notable property of the coolant is that it does not have a boiling point; it thermally decomposes (the molecular structure ‘cracks’) at a temperature of about 350°C. The operating temperature range for the fluid lies between –66°C and 300°C, so it is quite suitable for immersion cooling of servers.

The Volumetric Heat Capacity of EC-110 is about 1500 times higher than that of air as shown in Table 1, which clearly establishes it as a fluid with superior heating capacity. Although the value of Volumetric Heat Capacity is highest for water, it cannot be used as a direct cooling medium due to reasons discussed in section 1.

2.2. Server under study

The server chosen for study is a Third Generation Open Compute Server [8]. It consists of two Intel Xeon Microprocessors with a Thermal Design Power of 115 W each. However, for this study, the analysis is carried out at CPU powers of 65 W, 90 W, 115 W, and 150 W, to see if the cooling method is suitable for higher powers, and also to observe the trend. It has 4 DIMM (Dual In-Line Memory Module) blocks on each side of each server, making it a

Table 1
Comparison of Heat Capacities of Air and Dielectric Fluid.

Fluid	Heat Capacity (kJ/kg-K)	Density (kg/m ³)	Volumetric Heat Capacity (kJ/m ³ -K)
Air	1.01	1.225	1.237
Water	4.184	1000	4184
EC-110	2.1523	842.56	1813.44

total of 16 DIMMs. The chips have parallel plate copper heat sinks attached to them. The dimensions of server are 166.2 mm × 511 mm.

The server model has been developed in SolidWorks as shown in Fig. 1 and in further it is transferred to ANSYS Fluent for analysis. The DIMM blocks are assumed to be a single block, each generating no heat.

2.3. Height of the server

The height of the server, also known as the form factor, is taken in Rack Units denoted by U. 1U equals 44.45 mm. Another unit known as Open Rack Unit is used as the standard unit for server heights. One Open Rack Unit equals 48 mm. The standard conversion of rack units in mm is given in Table 2 [11].

The original form factor of the server is 2U. However, for this study, the form factor of the server has been reduced to 1 Open Rack Unit or 48 mm as shown in Fig. 2.

2.4. Volumetric flow rate and velocity calculation

The volumetric flow rate values for the fluid circulation are varied from 0.5 lpm to 2.5 lpm with increment of 0.5 lpm, based on previous experiments and they are the recommended flow rate values [7&10]. Table 3 shows the flow velocity values for different flow rates of cooling fluid and the calculation of flow velocity is done as shown below,

$$\text{Volume Flow Rate} = \text{Area} \times \text{Velocity} \quad (1)$$

$$\text{Velocity} = \frac{\text{Volume Flow Rate}}{\text{Area}}$$

$$\text{Area} = \text{Width} \times \text{Height} = 0.1662 \text{ m} \times 0.048 \text{ m} = 0.0079776 \text{ m}^2$$

$$\text{Volume Flow Rate} = 8.33 \times 10^{-6} \text{ m}^3/\text{s}$$

$$\text{Velocity} = \frac{8.33 \times 10^{-6} \text{ m}^3/\text{s}}{0.0079776 \text{ m}^2} = 0.0010445 \text{ m/s}$$

2.5. Reynold's number calculation

The Reynold's number is necessary to determine the type of flow (laminar, turbulent or transitional). For this, properties of the fluid such as density, velocity, dynamic viscosity and hydraulic diameter need to be calculated. The properties of the fluid at 30°C are determined and are presented in Table 4.

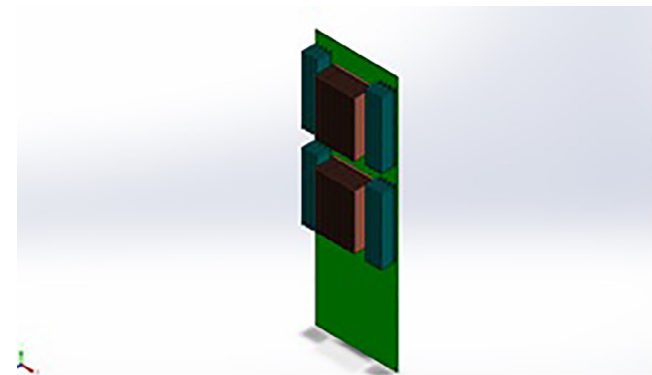


Fig. 1. Isometric View of Server in SolidWorks.

Table 2
Height of Server.

	Rack Unit (in)	Rack Unit (mm)	Open Rack Unit (mm)
1U	1.75	44.45	48
1.5U	2.625	66.675	72
2U	3.5	89.9	96

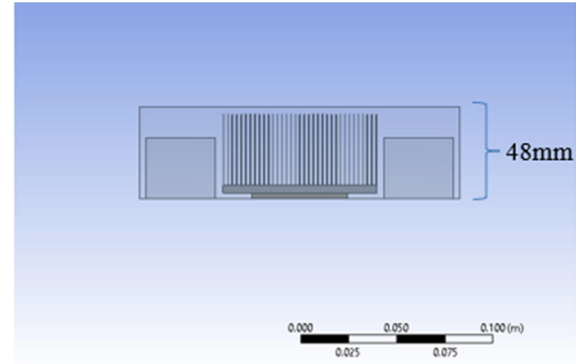


Fig. 2. Height of Server (1 Open Rack Unit).

Table 3
Velocity Calculation for Cooling Fluid.

Volume Flow Rate (lpm)	Volume Flow Rate (m³/s)	Velocity (m/s)
0.5	8.33×10^{-6}	0.001045
1	1.67×10^{-5}	0.002089
1.5	2.5×10^{-5}	0.003134
2	3.33×10^{-5}	0.004178
2.5	4.17×10^{-5}	0.005223

$$\text{Dynamic Viscosity } (\mu) = 0.00716 \text{ kg/ms}$$

$$\text{Density } (\rho) = 839.26 \text{ kg/m}^3$$

The formula for Hydraulic Diameter is given as,

$$D = \frac{2bh}{(b+h)} \quad (2)$$

where, b = width = 166.2 mm, h = height = 48 mm

$$D = \frac{2 \times 166.2 \text{ mm} \times 48 \text{ mm}}{(166.2 \text{ mm} + 48 \text{ mm})} = 74.4873 \text{ mm} = 0.074 \text{ m}$$

$$Re = \frac{\rho VD}{\mu} = \frac{839.26 \text{ kg/m}^3 \times 0.00104 \text{ m/s} \times 0.074 \text{ m}}{0.00716 \text{ kg/ms}} = 9.1195778 \quad (3)$$

where Re = Reynold's Number.

Similarly, the Reynold's Number is calculated at 25°C, 35°C, and 40°C. The highest value of Reynold's Number has been found to be 60.77 at 40°C inlet and 2.5 lpm flow rate as found from Table 5. Since $Re \ll 2300$, the flow is laminar and is modelled as laminar flow.

2.6. CFD modelling

Based on the value of Reynold's number, the flow is modelled as laminar flow. The governing differential equations for laminar modelling are as follows:

Continuity Equation:

Table 4Reynold's Number Values at 30°C, $\rho = 839.26 \text{ kg/m}^3$.

Flow rate (lpm)	Velocity (m/s)	Hydraulic Diameter (m)	Viscosity(kg/m-s)	Re
0.5	0.0010445	0.074487395	0.00716	9.1195778
1	0.0020891	0.074487395	0.00716	18.2400287
1.5	0.0031337	0.074487395	0.00716	27.36047961
2	0.0041783	0.074487395	0.00716	36.48093051
2.5	0.0052229	0.074487395	0.00716	45.60138142

Table 5Reynold's Number Values at 40°C, $\rho = 832.66 \text{ kg/m}^3$.

Flow rate (lpm)	Velocity (m/s)	Hydraulic Diameter (m)	Viscosity(kg/m-s)	Re
0.5	0.0010445	0.074487395	0.00533	12.15434959
1	0.0020891	0.074487395	0.00533	24.30986282
1.5	0.0031337	0.074487395	0.00533	36.46537606
2	0.0041783	0.074487395	0.00533	48.6208893
2.5	0.0052229	0.074487395	0.00533	60.77640254

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{V}) = 0 \quad (4)$$

Navier Stokes Equation:

$$\rho \left(\frac{\partial \mathbf{V}}{\partial t} + (\mathbf{V} \cdot \nabla) \mathbf{V} \right) = -\nabla P + \rho \mathbf{g} + \mu \nabla^2 \mathbf{V} \quad (5)$$

Energy Equation:

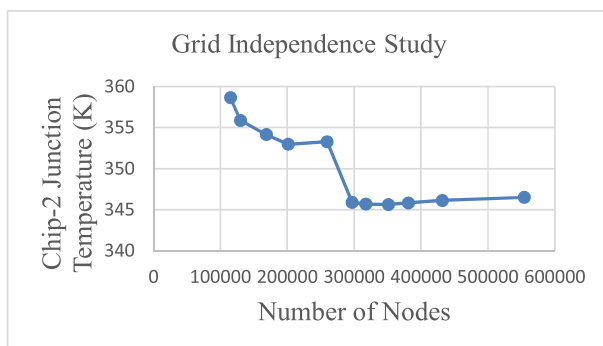
$$\frac{\partial (\rho C_p T)}{\partial x} + \nabla \cdot (\rho C_p T \mathbf{V}) = \nabla \cdot (k \nabla T) + S \quad (6)$$

Where ρ = density, ∇ = divergence, μ = Viscosity, P = Pressure, C_p = Specific Heat, T = temperature, k = thermal conductivity, S = Source Term

2.7. Grid independence study

A grid independence study has been carried out by parameterising the mesh size, and measuring the average chip junction temperature at Chip 2. The study has been carried out at a power of 65 W, inlet temperature of 30 °C, and flow rate of 0.5 lpm.

As is shown in the Fig. 3, once the node count reaches 296968, reducing the size further does not result in significant change in the value of the temperature, in other words, it remains almost constant. Hence, the node count of 296,968 is chosen to carry out the rest of the analysis, which gives the most accurate and computationally inexpensive results.

**Fig. 3.** Grid Independence Check.

2.8. Validation of CFD model

The CFD model is validated with a previously developed CFD model in ANSYS Icepack by Dandamudi [12]. For this, the fluid taken is EC-100, which was used in the previous model. The value of chip junction temperature at Chip 2 is determined at inlet temperature of 30 °C, and power of 65 W. Thus, the values of temperature are obtained with an accuracy of 10% as reported in Table 6.

3. Results and discussion

For the analysis, the range of powers on both the CPUs has been taken to be 65 W, 90 W, 115 W, and 150 W. For each value of power, the inlet temperature is varied from 25 °C to 40 °C at intervals of 5 °C. Further for each value of inlet temperature, the flow rate is varied from 0.5 lpm to 2.5 lpm. A total of 80 simulations have been performed on ANSYS Fluent. The output parameter of interest is the value of chip junction temperatures on Chip 1 and Chip 2. For safe operation of the server, the chip junction temperature should not exceed 85 °C [3].

3.1. Temperature contours

The temperature contours are obtained for all the simulations carried out. For conditions of 25°C inlet, 0.5 lpm flow rate, and 65 W power, the contours are shown in Fig. 4 & Fig. 5. The temperatures around Chip 2 are higher than those around Chip 1. This is due to the fact that when the fluid passes over Chip 1, it absorbs heat from the heat sink and its temperature rises. Thus, the fluid at higher temperature is unable to remove heat as effectively from Chip 2. This phenomenon is known as thermal shadowing.

3.2. Trends in values of chip junction temperatures

The values of chip junction temperatures are computed, and graphs of temperature vs flow rate are plotted at all the values of power, from Figs. 6–13. In Figs. 6–9, for a given power, the

Table 6

Validation of CFD Model.

Flow rate (lpm)	Temperature of present model (°C)	Temperature of previous model (°C)	Error (%)
0.5	69.51	67.39	3.14
1	58.59	61.25	−4.34
1.5	51.26	55.6	−7.8

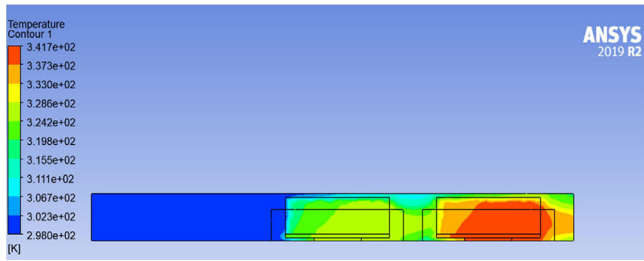


Fig. 4. Contours on YZ Plane at 25°C, 0.5 lpm, 65 W.

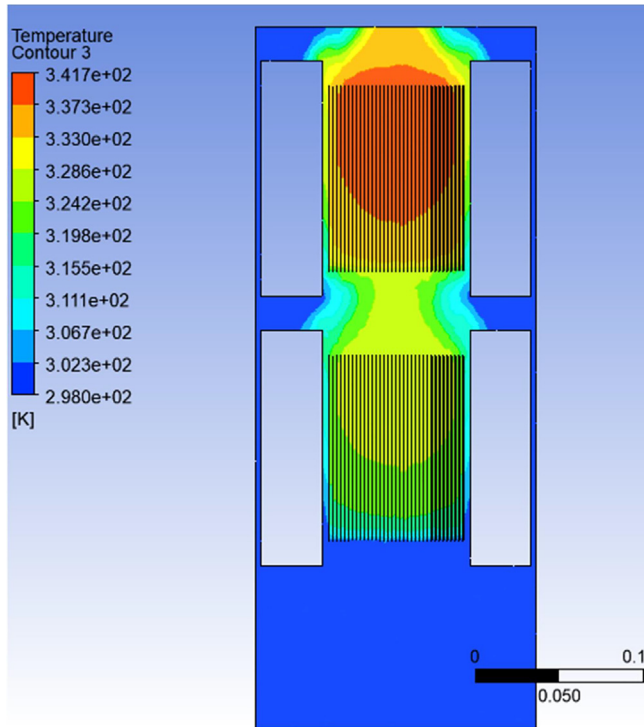


Fig. 5. Contours on XY Plane at 25°C, 0.5 lpm, 65 W.

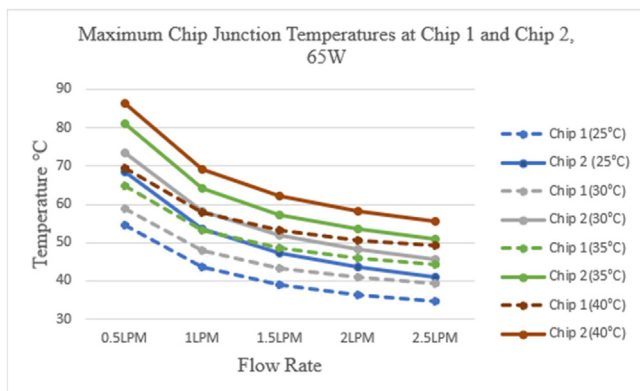


Fig. 6. Temperature vs Flow Rate at 65 W.

variation of the chip junction temperatures at various values of the inlet temperatures and flow rates is plotted. In Figs. 10–13, for a given inlet temperature, the variation of the chip junction temperatures at various values of power and flow rates are plotted.

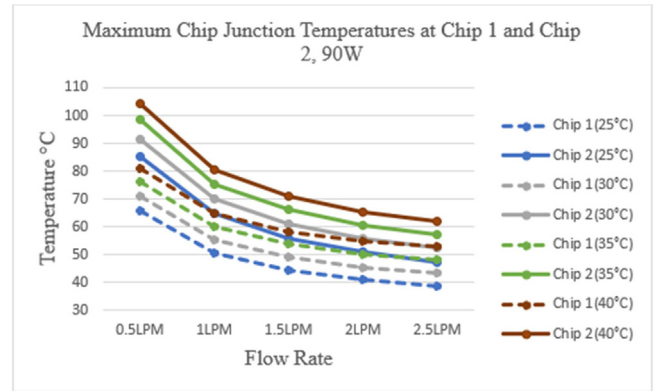


Fig. 7. Temperature vs Flow Rate at 90 W.

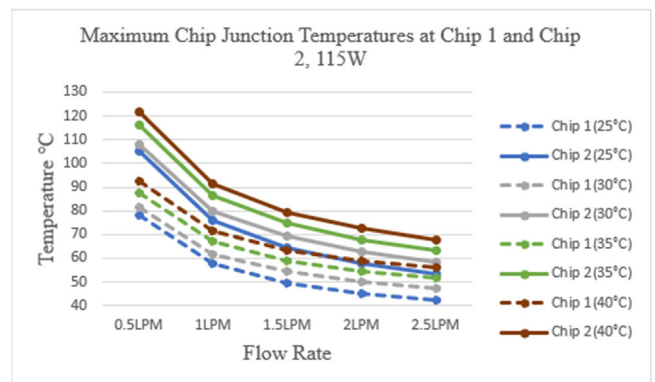


Fig. 8. Temperature vs Flow Rate at 115 W.

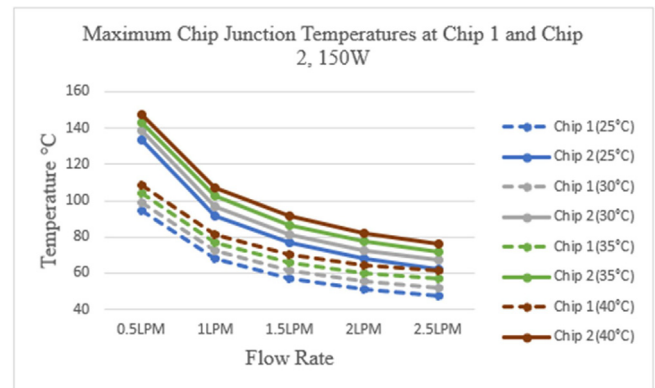


Fig. 9. Temperature vs Flow Rate at 150 W.

Overall, there is a steep fall in the chip junction temperature values, when the flow rate is increased from 0.5 lpm to 1 lpm. The fall in temperature becomes less steep for subsequent increases in the flow rate values. In fact, the change of temperature values is almost unchanged when the flow rate is increased from 2 lpm to 2.5 lpm. For a power of 65 W, it is possible to maintain both the chips at temperatures <85°C, even for a fluid inlet temperature of 40°C, and flow rate of 1 lpm as is visible in Fig. 6.

As expected, from Fig. 7, when the power is increased to 90 W, the chip junction temperatures experience an overall increase. However, even in this case, the server can be maintained at safe operating conditions when the inlet temperature is 40°C and flow rate is around 1 lpm. The chip junction temperature at chip 2 is

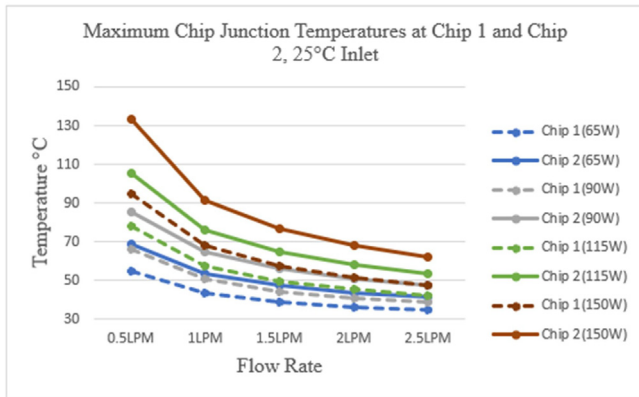


Fig. 10. Temperature vs Flow Rate at 25 °C.

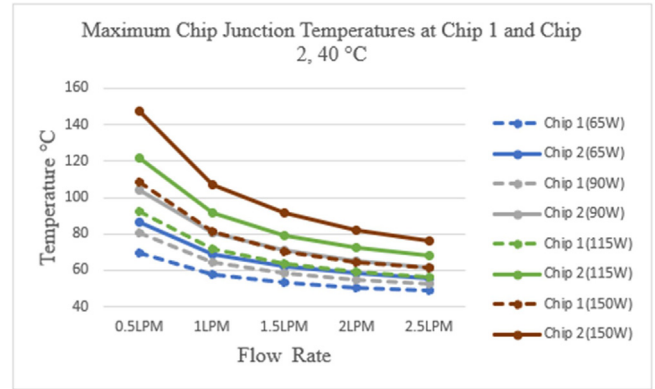


Fig. 13. Temperature vs Flow Rate at 40 °C.

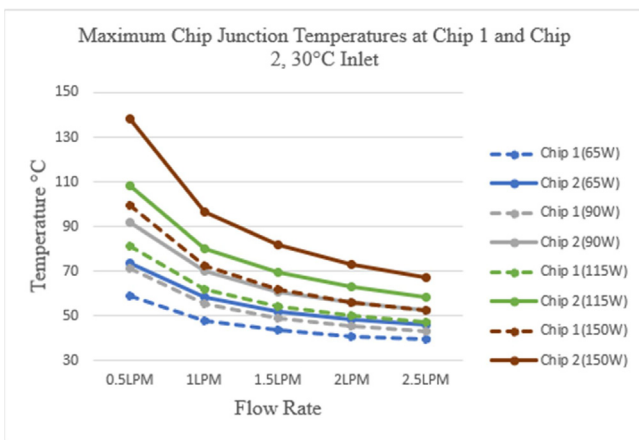


Fig. 11. Temperature vs Flow Rate at 30 °C.

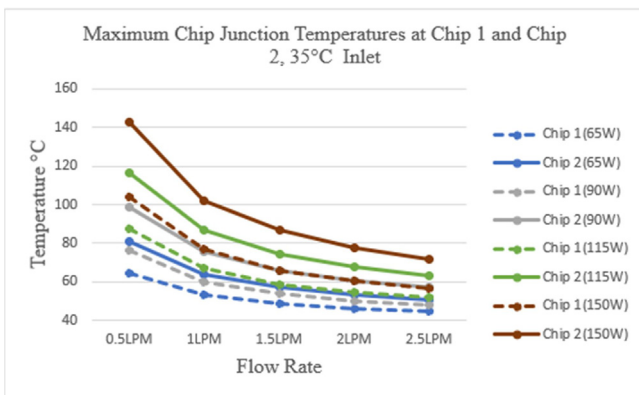


Fig. 12. Temperature vs Flow Rate at 35 °C.

always higher than that at chip 1. However, as the flow rate increases, the difference in the values decreases.

As the power is increased to 115 W, the overall values of chip junction temperatures further increase, exceeding 120°C at a flow rate of 0.5 lpm, as seen in Fig. 8.

As shown in Fig. 9, at 150 W and 0.5 lpm, the chip junction temperature for chip 2 reaches >120°C, even when the inlet temperature is 25°C. However, at higher values of flow rate, it is possible to maintain the chip junction temperature at safe levels, even for an inlet temperature of 40°C.

At inlet temperature of 25 °C, at lower flow rates, the chip junction temperature can even exceed 130 °C at a flow rate of 0.5 lpm. This temperature value drops to about 90 °C when the flow rate is doubled. Similar trends are observed at higher values of inlet temperatures as is visible in Figs. 10–13. At lower values of power, the chip junction temperature curve remains relatively flat when compared to the curve at higher values of CPU power, in which case, the curve falls rather steeply.

Overall, as the power increases from 65 W to 150 W, the chip junction temperatures increase for the same value of flow rate and fluid inlet temperature. Similarly, as the fluid inlet temperature increases from 25 °C to 40 °C, the value of chip junction temperature increases for the same value of power, and flow rate. At higher values of power and fluid inlet temperature, higher flow rates are required to maintain the chip at safe temperatures.

The obtained data set contains numerous values of chip junction temperatures at different power, flow rates and inlet temperature which makes a hideous task in calculating the desired values at specific parameters. In order to remove this complexity, regression analysis has been performed further for determining the optimum temperature values by formulating an equation.

3.3. Regression model

Regression analysis is a powerful method which helps in determining the closeness of actual values in practice. A generalized equation for chip 1 and chip 2 junction temperatures is created using multiple regression in Minitab software. Different values for power (W), inlet temperature (°C) and flow rate (lpm) are taken as input into the developed regression equation and chip junction temperature (°C) are determined. The values of chip junction temperature obtained from the equation are found to be within 5–10% error limitations when compared to the value obtained from ANSYS.

3.4. Regression equations for chip 1 & chip 2

$$\begin{aligned} \text{Chip 1 Junction Temperature (}^{\circ}\text{C)} \\ = 24.74 + 0.2545x_1 + 0.938x_2 - 15.307x_3 \end{aligned} \quad (7)$$

$$\begin{aligned} \text{Chip 2 Junction Temperature (}^{\circ}\text{C)} \\ = 32.28 + 0.4133x_1 + 1.022x_2 - 22.64x_3 \end{aligned} \quad (8)$$

where,

Table 7
Model Summary of Chip-1 & Chip-2.

Chip	S (Standard Error)	R-Squared	R-Squared (adjusted)	R-Squared (Predicted)
Chip 1	5.89219	86.35%	85.81%	84.52%
Chip 2	8.75943	86.26%	85.72%	84.40%

$$x_1 = \text{Power (W)}$$

$$x_2 = \text{Inlet Temperature (}^{\circ}\text{C)}$$

$$x_3 = \text{Flow Rate (lpm)}$$

In order to test the goodness of regression fit, R-squared and adjusted R-squared indicators are used as presented in Table 7.

The value of R-Squared and adjusted R-squared for Chip-1 is found to be 86.35 % and 85.81 % whereas, for chip 2 the same values are found to be 86.26 % and 85.72 % respectively which is a good approximation for the regression fit. In general, R-squared value provides a measure to how well the regression model corresponds with the observed data. In the present study, a R-squared value of 86.35% indicates that 86.35% of the data fits with the regression model successfully.

4. Conclusion

In this study, a conjugate heat transfer analysis is performed on a commercially used data server with the dielectric fluid EC-110, to study the chip junction temperature. The following results are observed from the present work,

- At 1.5 lpm and at inlet temperature of 25°C, the fluid is capable to maintain the chip junction temperature below threshold value; even when the CPUs are operated at 150 W.
- At the inlet temperature of 40°C, the fluid needs to be supplied at 2 lpm for the safe chip junction temperature (<85°C) while the CPUs are supplied with 150 W.
- The developed regression model is found to capture the variation of the chip junction temperature with input power, fluid inlet temperature, and flow rate.

CRediT authorship contribution statement

Aditya Chhetri: Conceptualization, Methodology. **Devendra Kashyap:** Investigation, Methodology. **Arvind Mali:** Writing – original draft. **Chaitanya Agarwal:** Data curation. **Caroline Ponraj:** Validation. **N. Gobinath:** Project administration, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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